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System Characterization - Drone Contral System

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1. Main Dashboard (The Operational Overview)

Purpose: To provide a high-level, real-time snapshot of the entire fleet's status, allowing managers to identify bottlenecks or emergencies instantly.

- **Layout:**
  - **Header:** Global search and system notifications.
  - **Top Row (KPI Cards):** Four summary cards: "Total Drones," "Active Missions," "Drones in Maintenance," and "Active Operators."
  - **Center Left (Live Map):** A map interface showing geolocated icons of active drones.
  - **Center Right (Fleet List):** A scrollable table showing real-time status.
- **Data Fields & Tables:**
  - **Table: Drones** – Displaying Drone_ID, Model_Name, Current_Battery_Level, Current_Latitude, Current_Longitude.
  - **Table: Missions** – Displaying Mission_ID, Mission_Status (e.g., In-Progress, Scheduled).
  - **Table: Operators** – Displaying Operator_Name, Certification_Level, Current_Assignment.
  - **Table: Status_Codes** – Linked via Status_ID (e.g., 1 = Flying, 2 = Idle, 3 = Charging, 4 = Maintenance).
- **Actions/Buttons:**
  - **[View Live Feed]:** Opens a modal with a camera stream for a specific drone.
  - **[Emergency Land]:** A high-priority button to force a drone to land.
  - **[Filter Map]:** Filter by drone model or operator.

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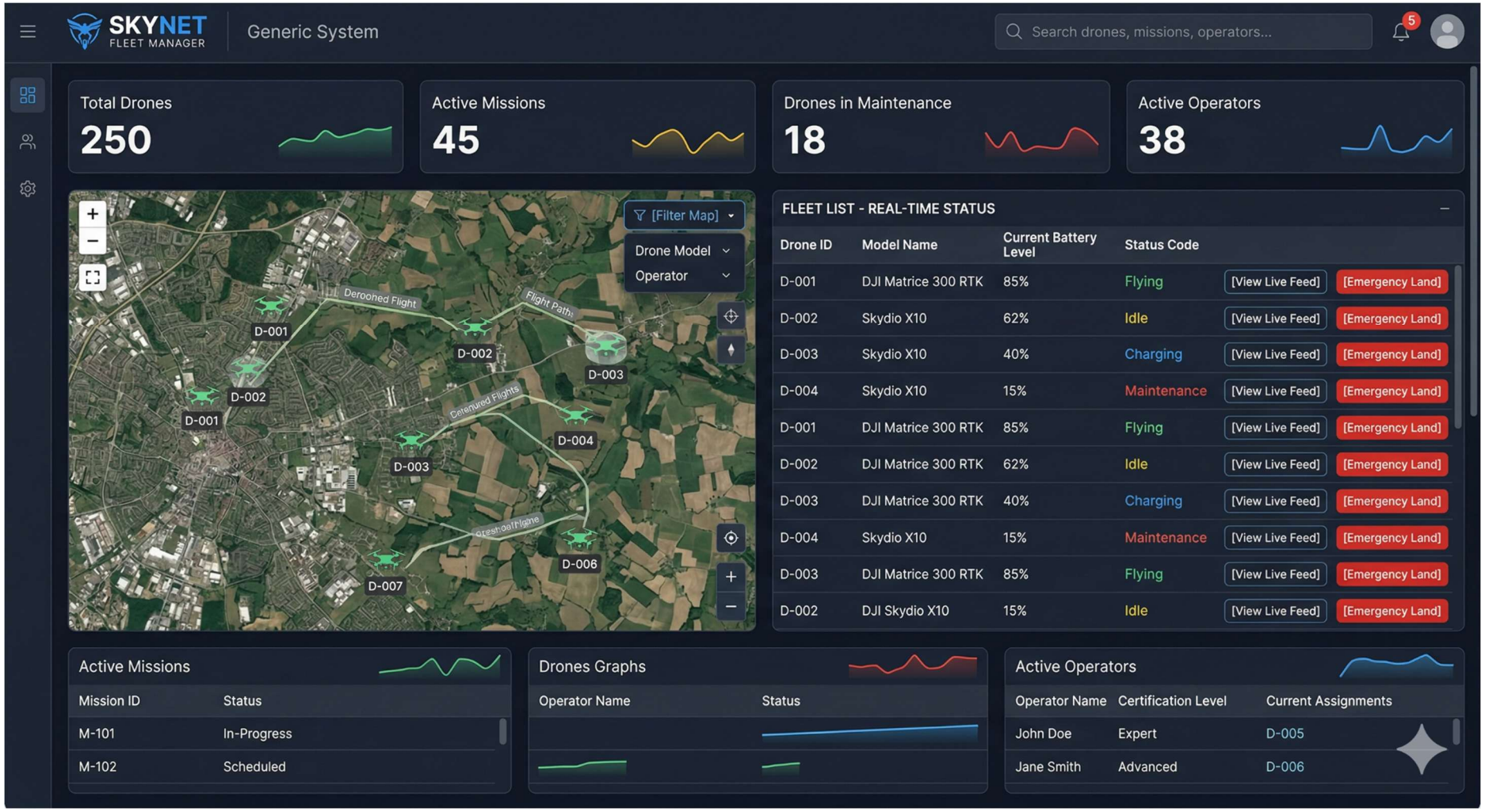
Example Screen - Main Dashboard



2. Mission Setup & Scheduling (The Resource Allocator)

Purpose: A transactional screen used to plan future flights. This screen is critical for your database as it handles many-to-many relationships and foreign key constraints.

- **Layout:**
  - **Left Pane (Form):** A step-by-step wizard to input mission details.
  - **Right Pane (Availability Checker):** A dynamic calendar view showing when drones and operators are free to prevent double-booking.
- **Data Fields & Tables:**
  - **Table: Missions (Input fields):** Mission_Title, Start_Time, End_Time, Target_Location_Coords, Mission_Type (e.g., Survey, Delivery, Inspection).
  - **Table: Drones (Selection):** Dropdown filtered by Availability_Status = 'Ready'.
  - **Table: Sensors/Payloads (Selection):** Sensor_ID, Sensor_Type (e.g., LIDAR, Thermal, RGB), Compatibility_Check.
  - **Table: Operators (Selection):** Dropdown filtered by License_Expiry_Date > Today.
- **Actions/Buttons:**
  - **[Check Compatibility]:** A logic check to ensure the selected sensor can be mounted on the selected drone.
  - **[Schedule Mission]:** Creates a new record in the Missions table and updates the Drone_Status to 'Reserved'.
  - **[Cancel]:** Discards inputs and returns to the dashboard.

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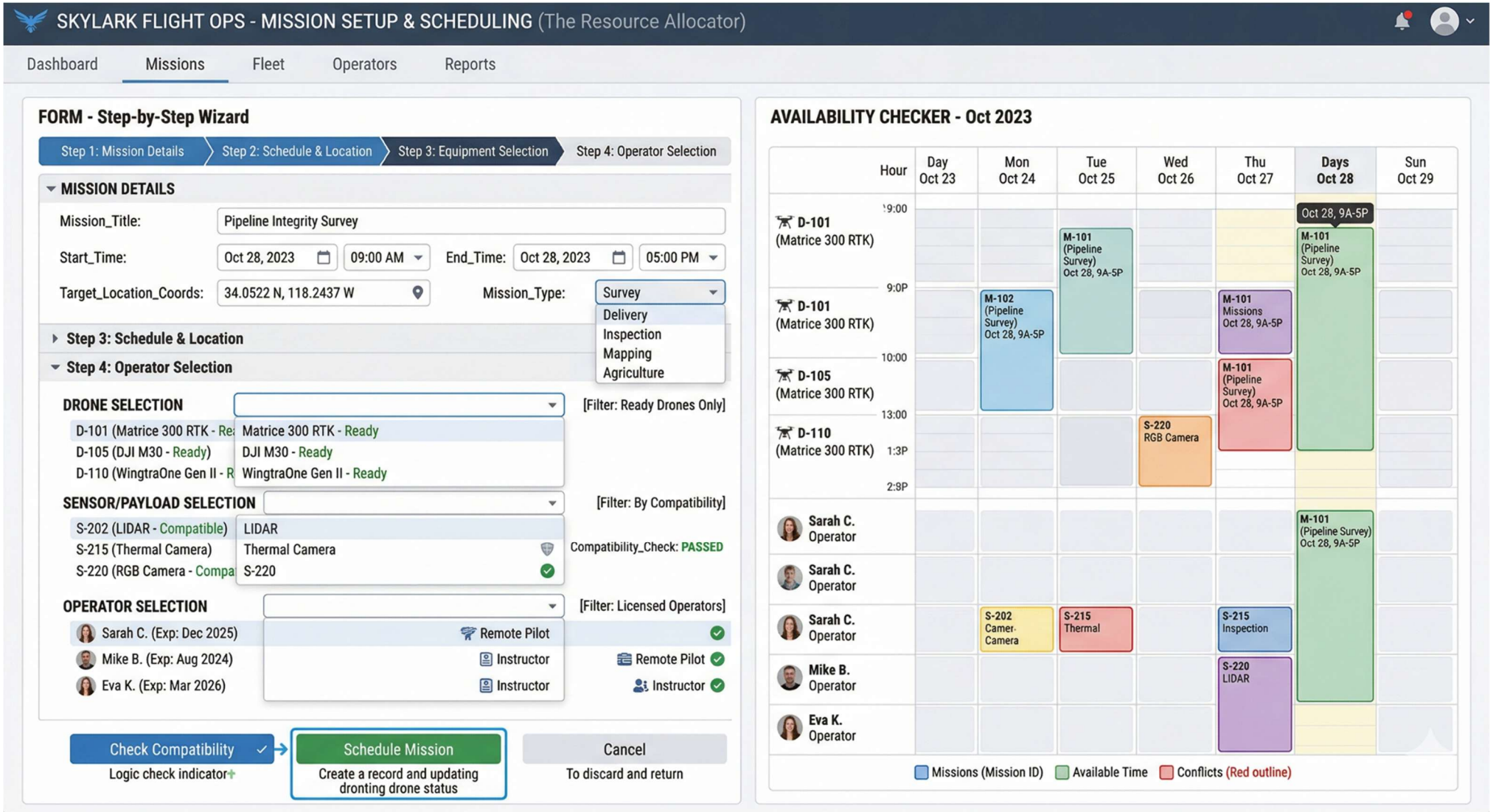
Example Screen - Mission Setup & Scheduling



3. Maintenance Hangar (The Health Monitor)

Purpose: To manage the lifecycle of the hardware. This allows the user to track wear and tear and schedule repairs based on flight hours.

- **Layout:**
  - **Grid Layout:** "Health Cards" for every drone in the fleet.
  - **Progress Bars:** Visual indicators for battery health, motor hours, and days until the next scheduled service.
  - **Bottom Section:** A history log of completed repairs.
- **Data Fields & Tables:**
  - **Table: Drones:** Total_Flight_Hours, Purchase_Date, Last_Service_Date.
  - **Table: Maintenance_Logs:** Log_ID, Drone_ID (FK), Technician_Name, Work_Performed_Description, Parts_Replaced, Maintenance_Cost.
  - **Table: Sensors:** Calibration_Status (Boolean), Firmware_Version.
- **Actions/Buttons:**
  - **[Log Maintenance]:** Opens a form to add a new entry to the Maintenance_Logs table.
  - **[Mark as Out-of-Service]:** Updates Drone_Status to 'Maintenance', making it unselectable in the Mission Setup screen.
  - **[Update Firmware]:** Triggers a (simulated) firmware update record.

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Example Screen - Maintenance Hangar

4. Flight Alerts Log (The Safety Audit)

Purpose: A historical and real-time ledger of system warnings. This screen is vital for post-flight analysis and incident reporting.

- **Layout:**
  - **Top Section:** Filters for "Severity" (Critical, Warning, Info) and "Date Range."
  - **Main Body:** A dense, chronological list of alert events.
  - **Sidebar:** An "Alert Details" pane that populates when a specific row is clicked.
- **Data Fields & Tables:**
  - **Table: Alerts:** Alert_ID, Timestamp, Severity_Level (1-3), Alert_Message (e.g., "Wind Speed Exceeded 30 knots").
  - **Table: Flights (Relationship):** Flight_ID (FK) to link the alert to a specific mission.
  - **Table: Telemetry_Snapshots:** Latitude_at_Alert, Longitude_at_Alert, Altitude_at_Alert, Battery_at_Alert.
- **Actions/Buttons:**
  - **[Export CSV/PDF]:** Generates a report for regulatory compliance (e.g., FAA reporting).
  - **[Acknowledge Alert]:** A boolean toggle (is_resolved) to show that a manager has reviewed the warning.
  - **[View Flight Path]:** Navigates to a map view of the specific flight where the alert occurred.

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Example Screen - Flight Alerts Log